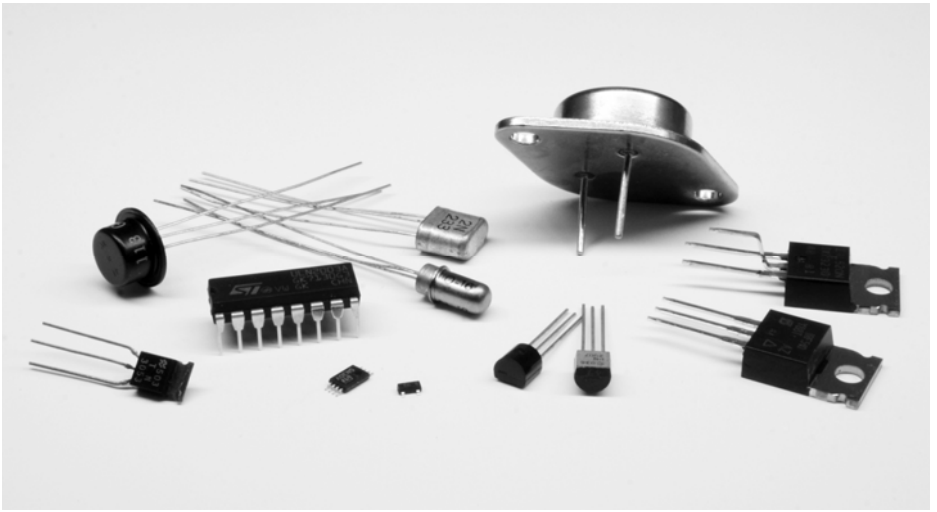


Figure 12-6. *Transistors*

There are two types of transistors that we are mostly concerned with. The NPN transistor, such as the 2N3904, is capable of sinking current in a circuit, allowing for current to flow from the collector pin to the emitter pin whenever current is applied to the base pin. Occasionally, we will need to use a PNP transistor, like the 2N3906, which reverses the direction of the NPN, sourcing current rather than sinking it, allowing for current to flow from the emitter pin to the collector pin whenever current is applied to the base pin. The concepts of sourcing and sinking were explained more thoroughly in Chapter 5, but Figure 12-7 provides two example schematics for how this works with NPN and PNP bipolar transistors.

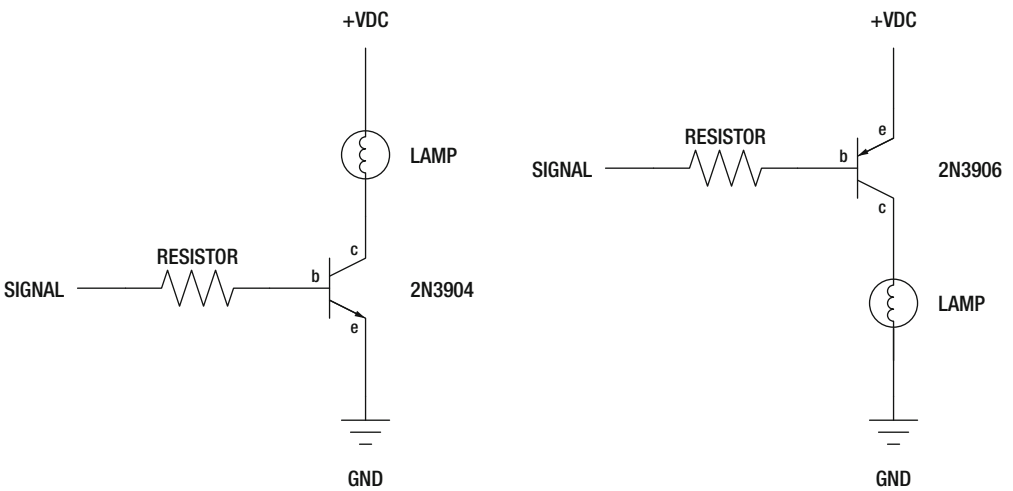


Figure 12-7. *NPN and PNP bipolar transistors*